

IVF / Fertility Clinic Management System

Patient

Activity Diagrams & Sequence Diagrams

Software Engineering – Laboratory Deliverable (Lab 06)

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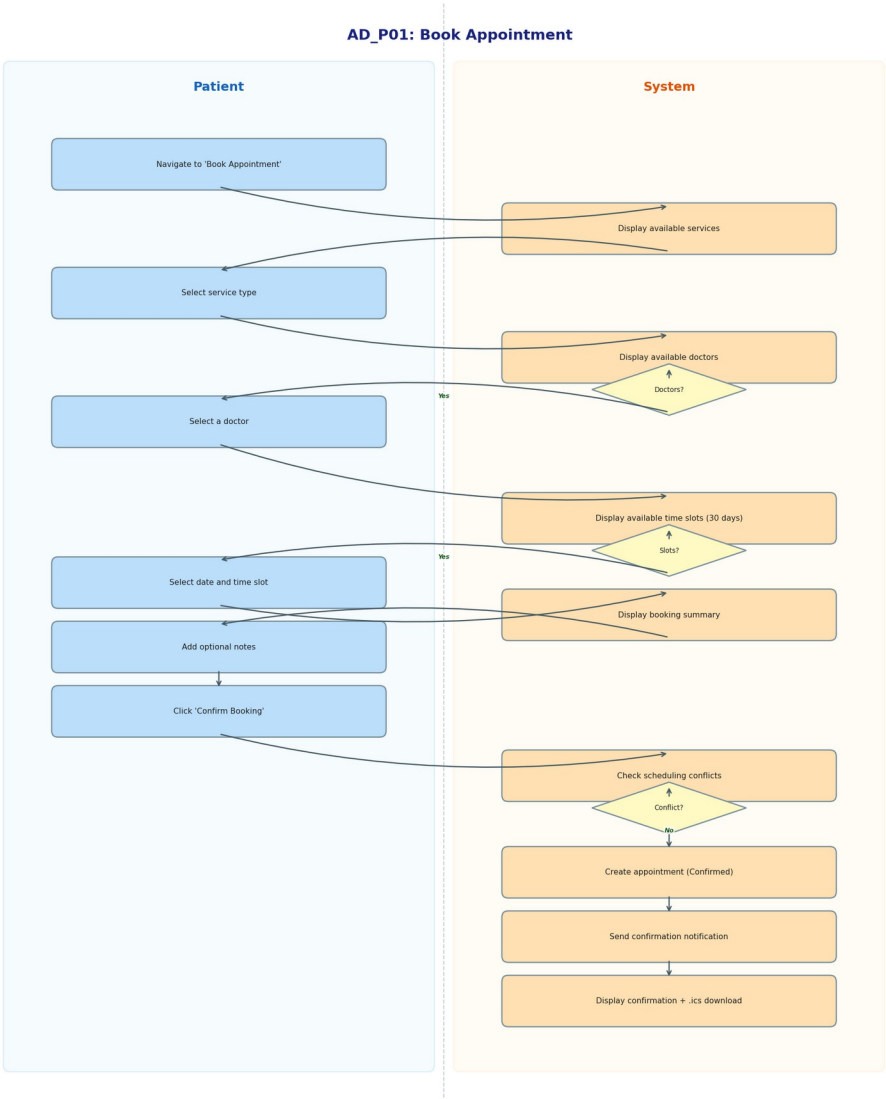
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1. Activity Diagrams

The following activity diagrams model the workflow of each Patient use case with Patient/System swimlanes.

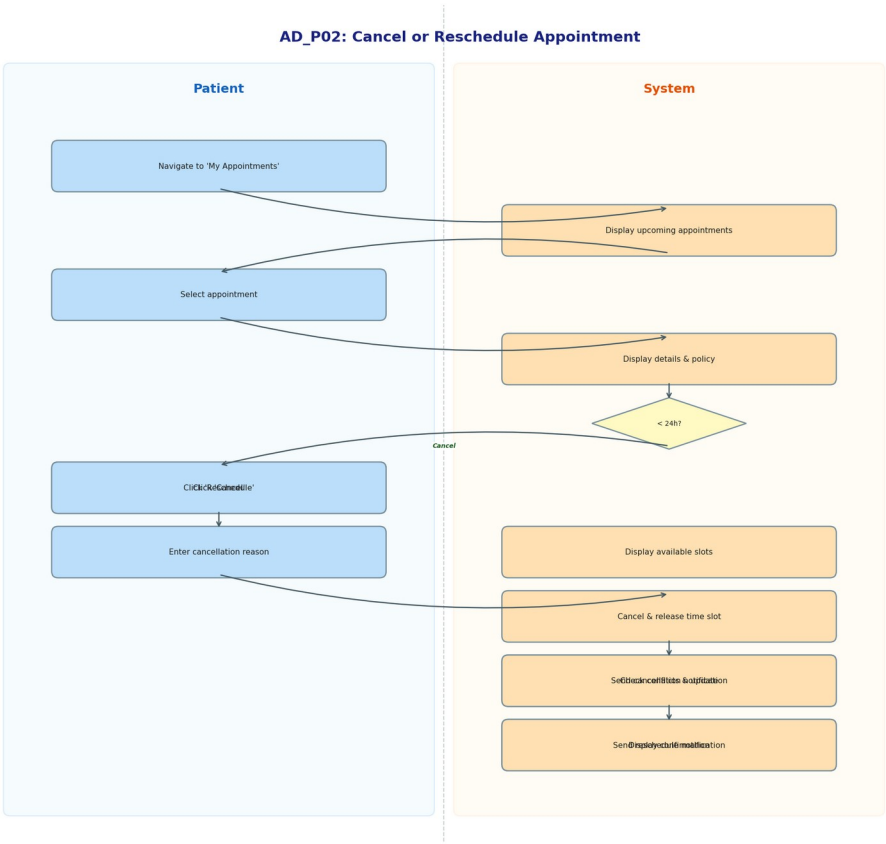
AD_P01: Book Appointment

Prerequisites: None. Shows service/doctor/slot selection, conflict check, confirmation.



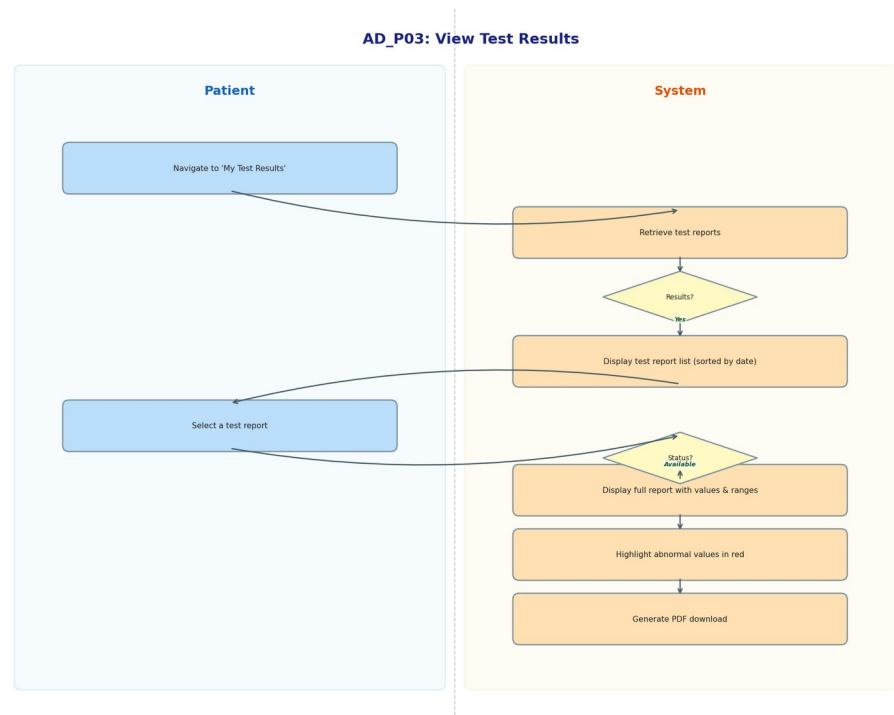
AD_P02: Cancel or Reschedule Appointment

Prerequisites: AD_P01. Shows cancel and reschedule paths with 24h policy.



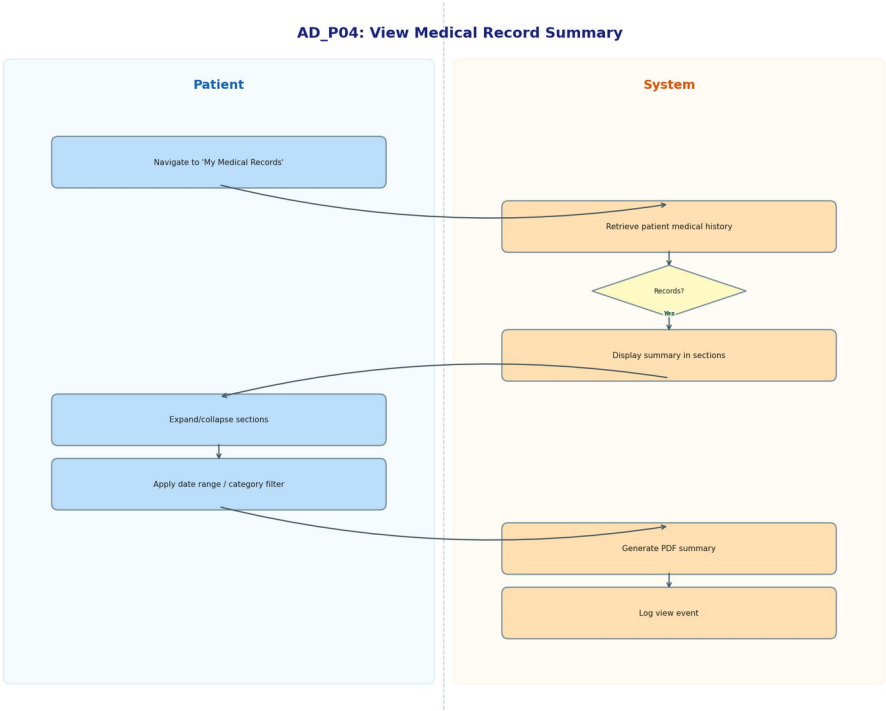
AD_P03: View Test Results

Prerequisites: AD_P01. Shows viewing released results, pending statuses, PDF download.



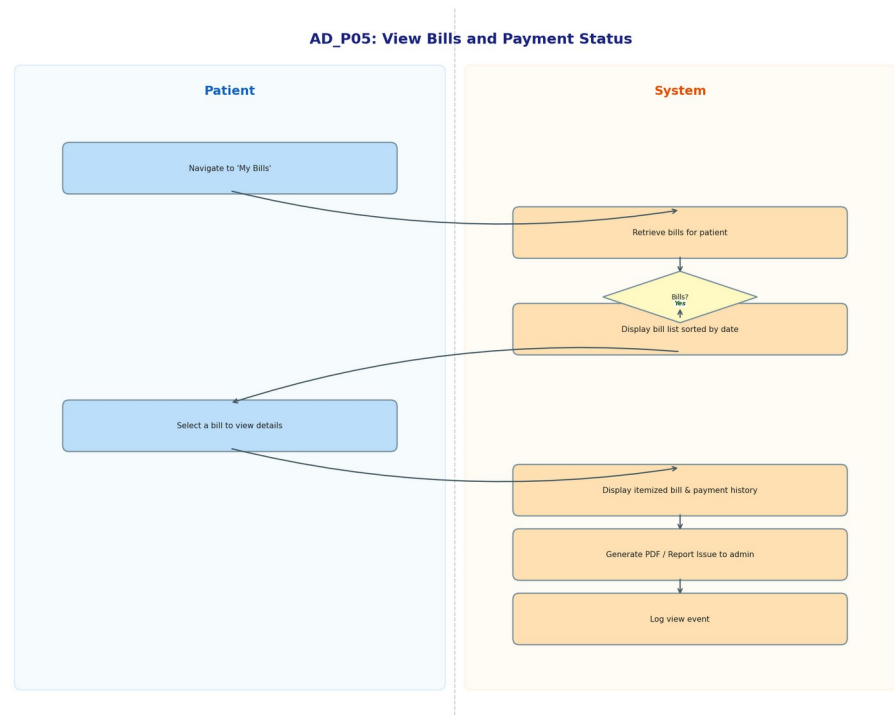
AD_P04: View Medical Record Summary

Prerequisites: AD_P01. Shows viewing medical history with filtering and PDF export.



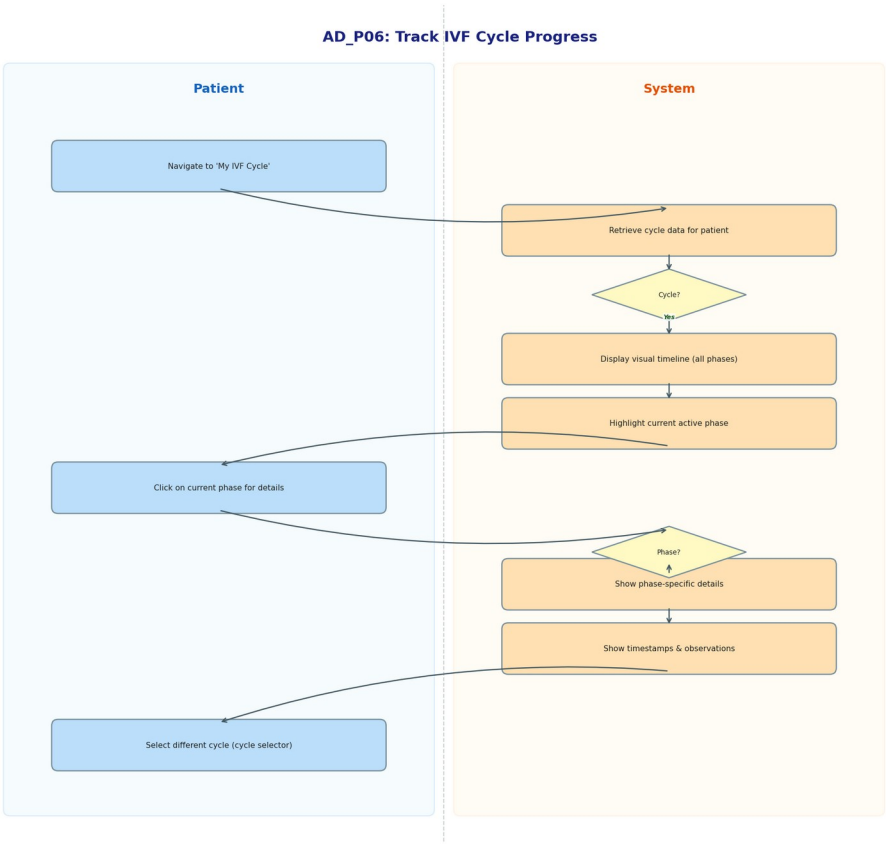
AD_P05: View Bills and Payment Status

Prerequisites: AD_P01. Shows viewing bills, payment history, reporting issues.



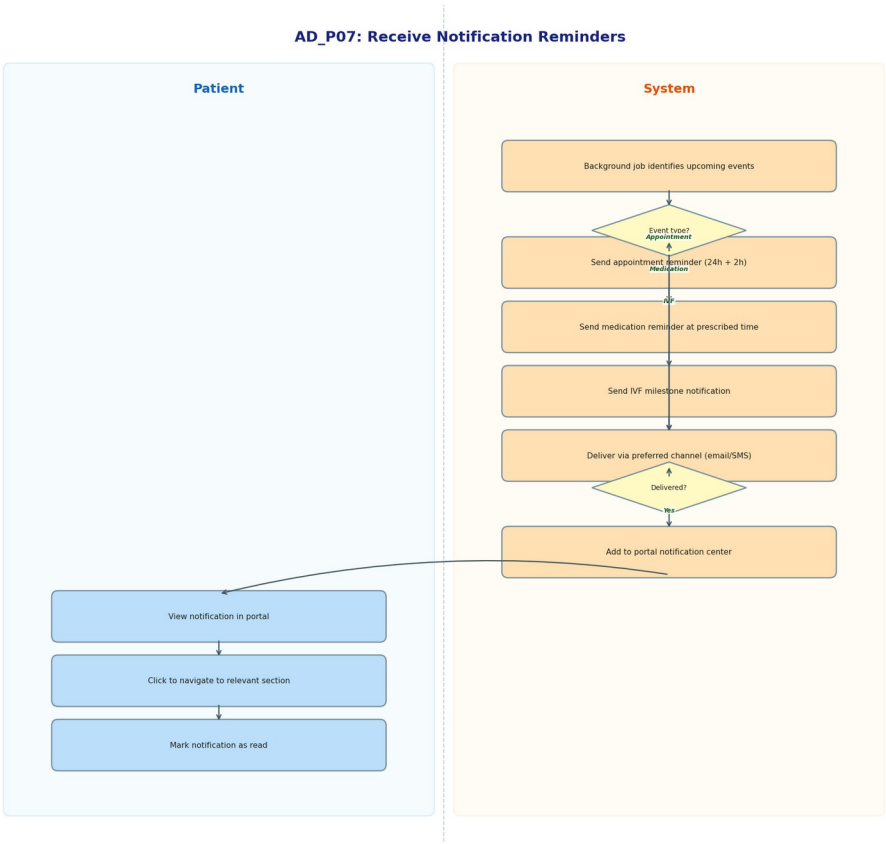
AD_P06: Track IVF Cycle Progress

Prerequisites: AD_P01. Shows monitoring IVF timeline and phase details.



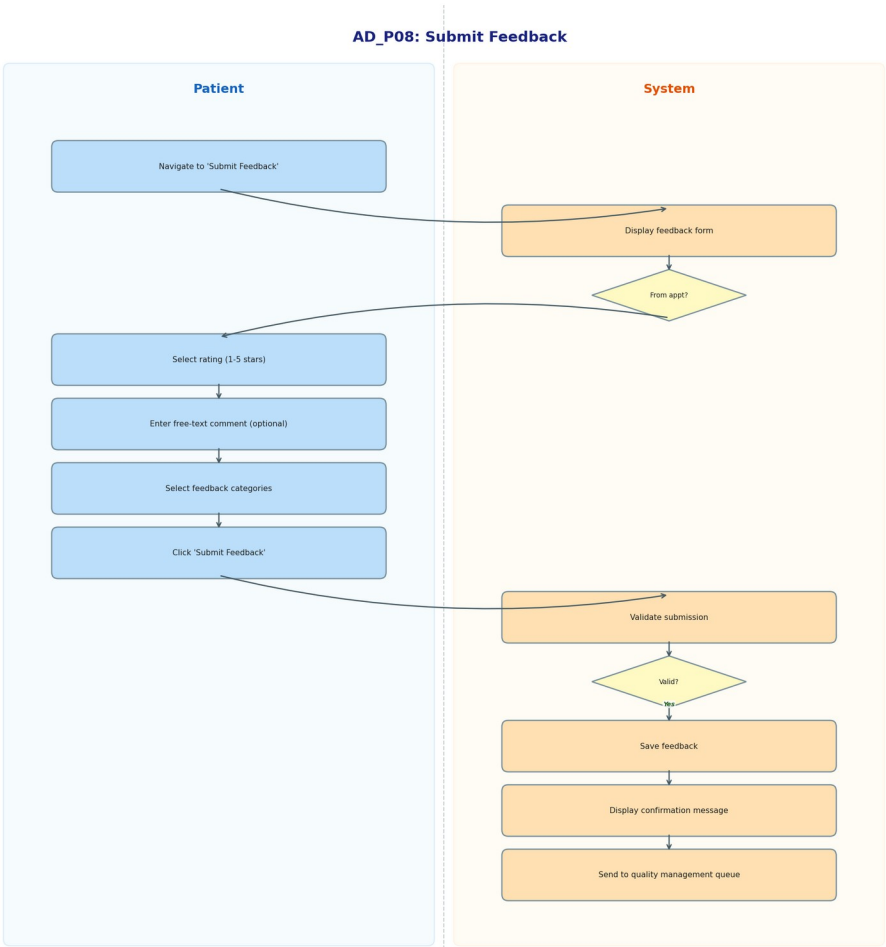
AD_P07: Receive Notification Reminders

Prerequisites: AD_P01, AD_P06. Shows system notification workflow with retry.



AD_P08: Submit Feedback

Prerequisites: AD_P01 (for appointment-linked). Shows feedback submission with validation.

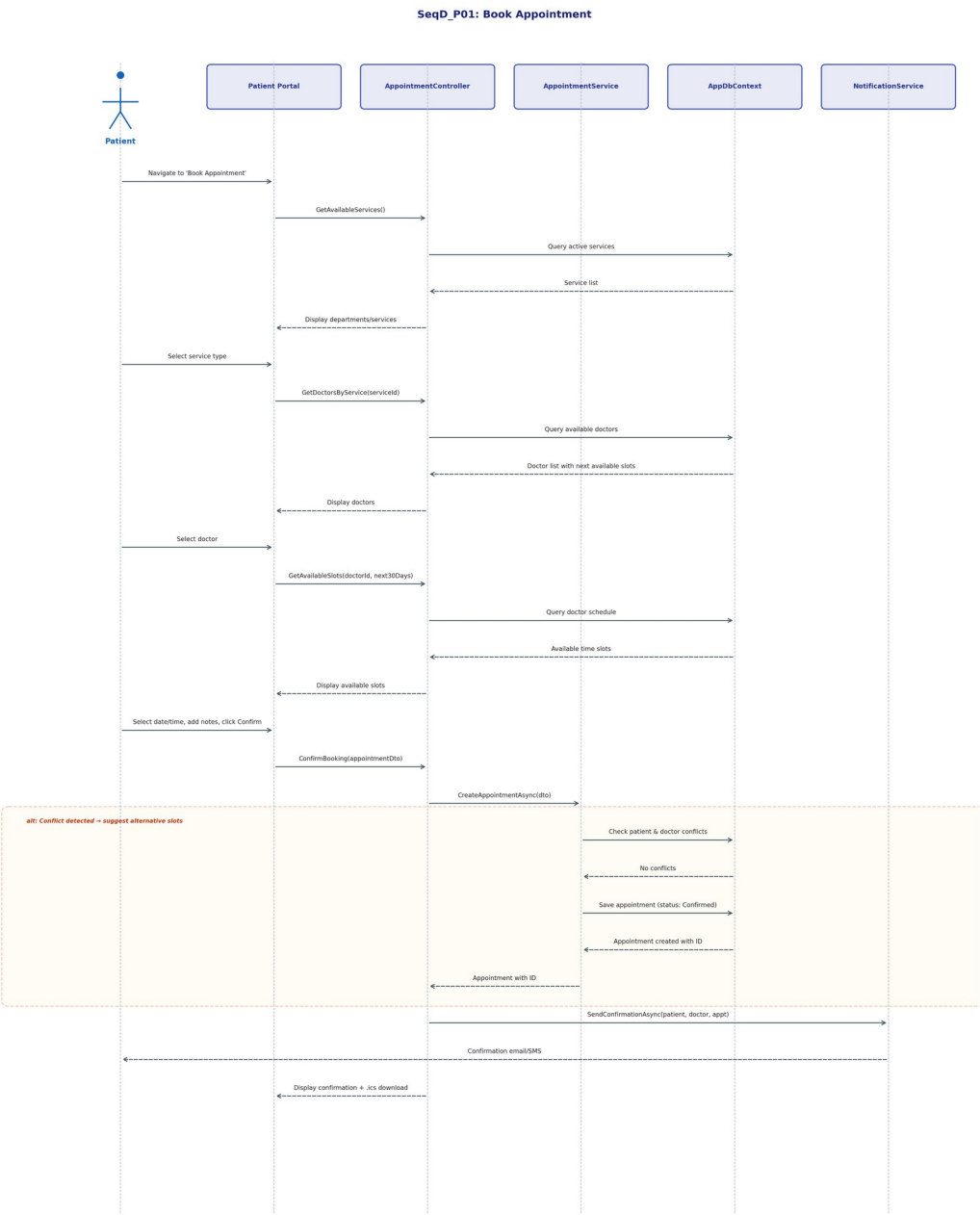


2. Sequence Diagrams

The following sequence diagrams show interactions between the Patient and system components over time.

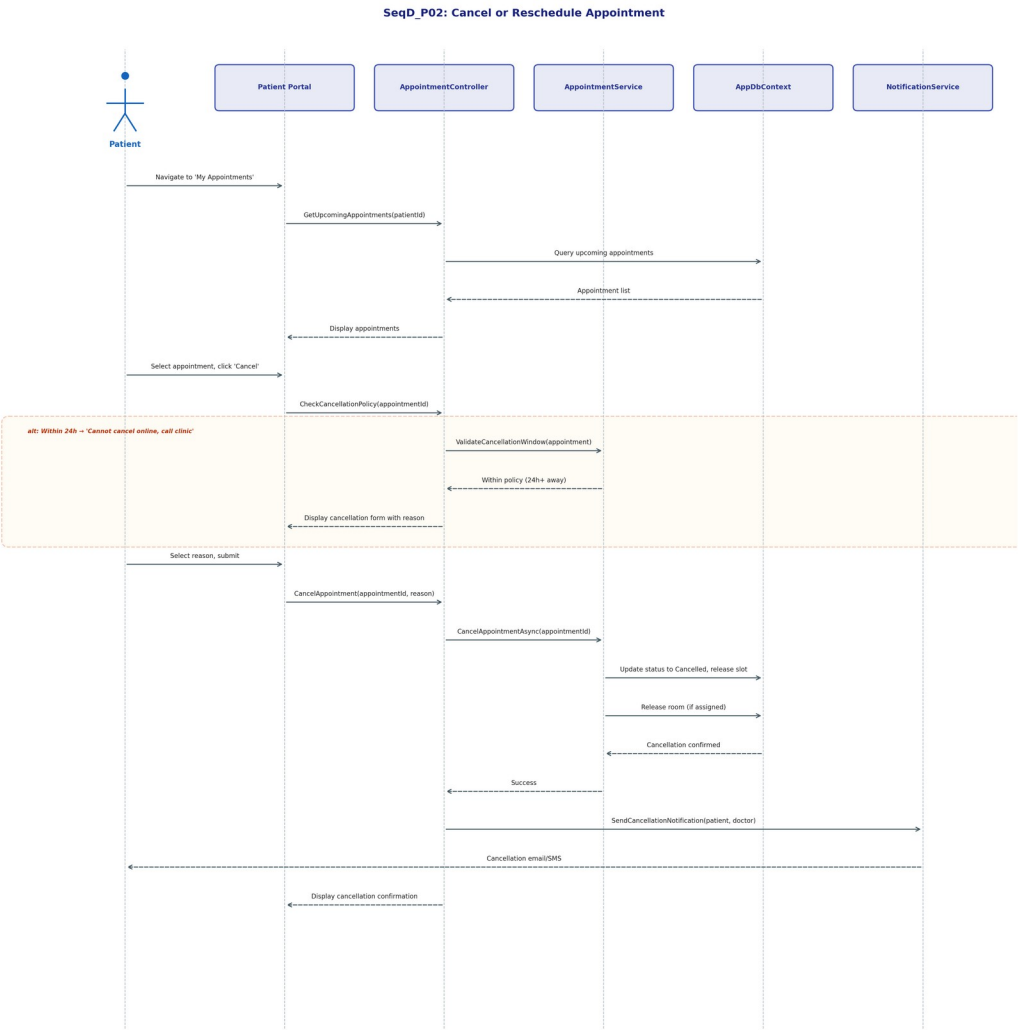
SeqD_P01: Book Appointment

Patient → Portal → AppointmentController → AppointmentService → AppDbContext → NotificationService



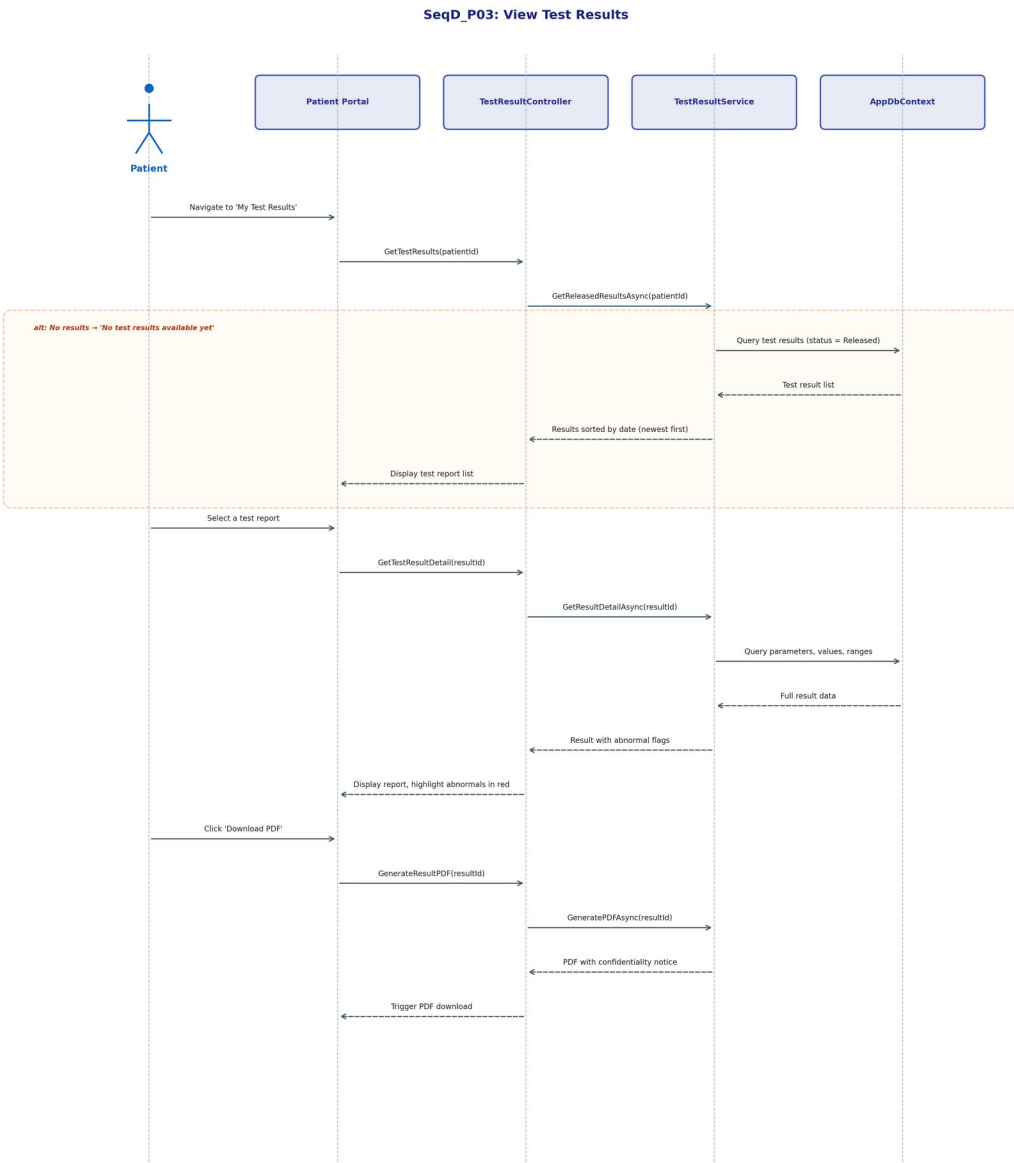
SeqD_P02: Cancel or Reschedule Appointment

Cancellation with policy validation, status update, slot/room release, notification.



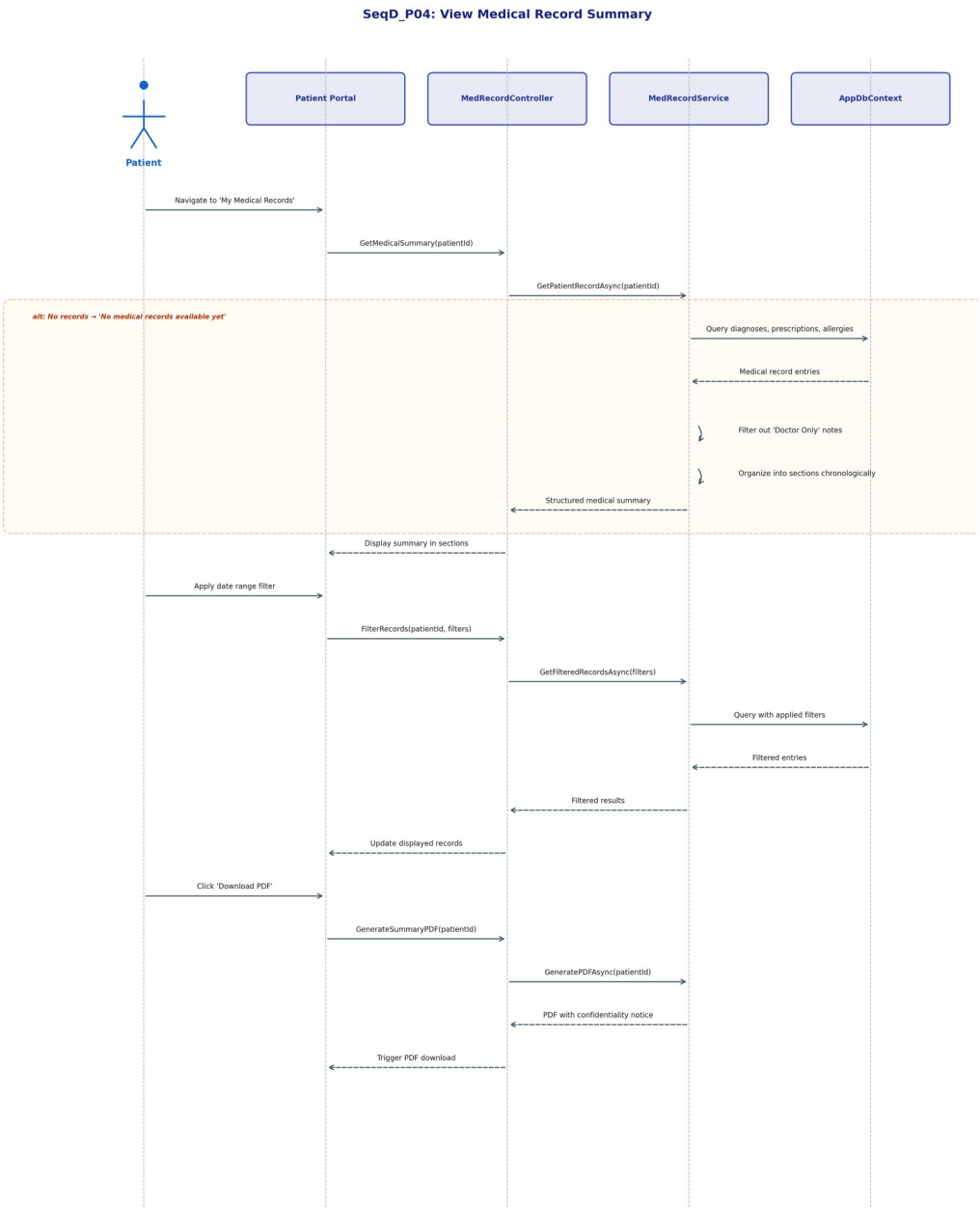
SeqD_P03: View Test Results

Retrieving released results, displaying abnormal flags, PDF generation.



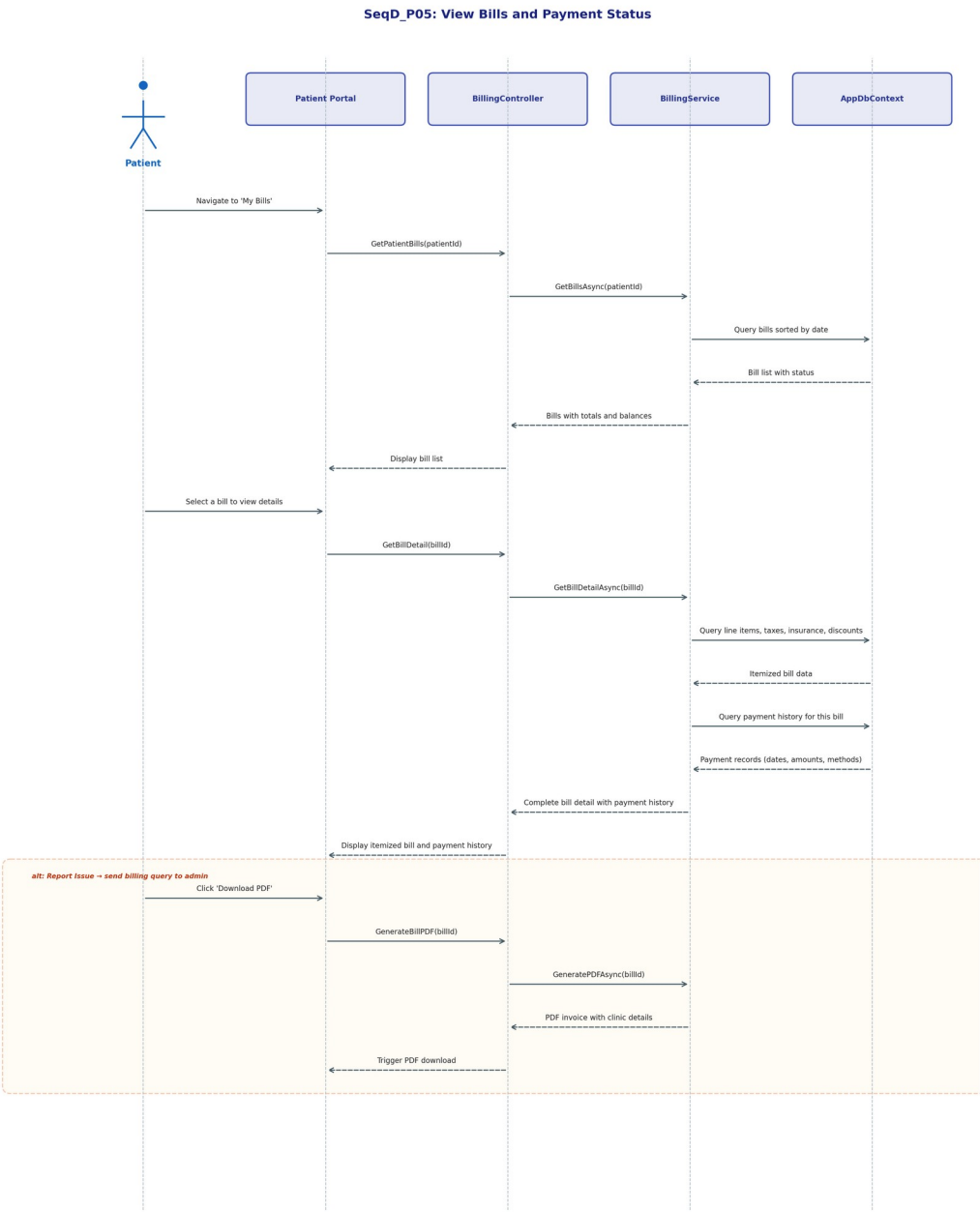
SeqD_P04: View Medical Record Summary

Retrieving medical history, filtering, applying date range, PDF export.



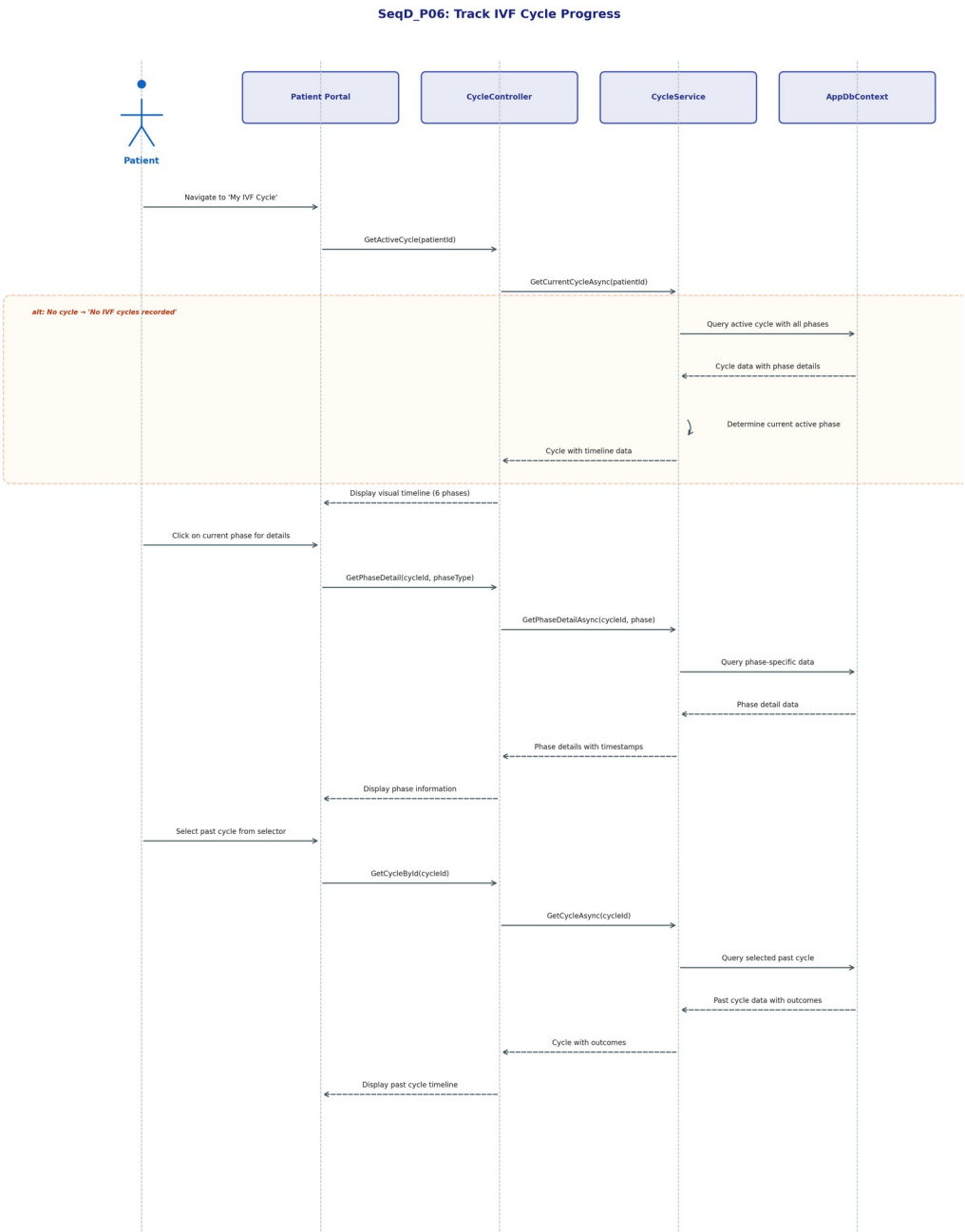
SeqD_P05: View Bills and Payment Status

Retrieving bills, itemized details, payment history, Report Issue flow.



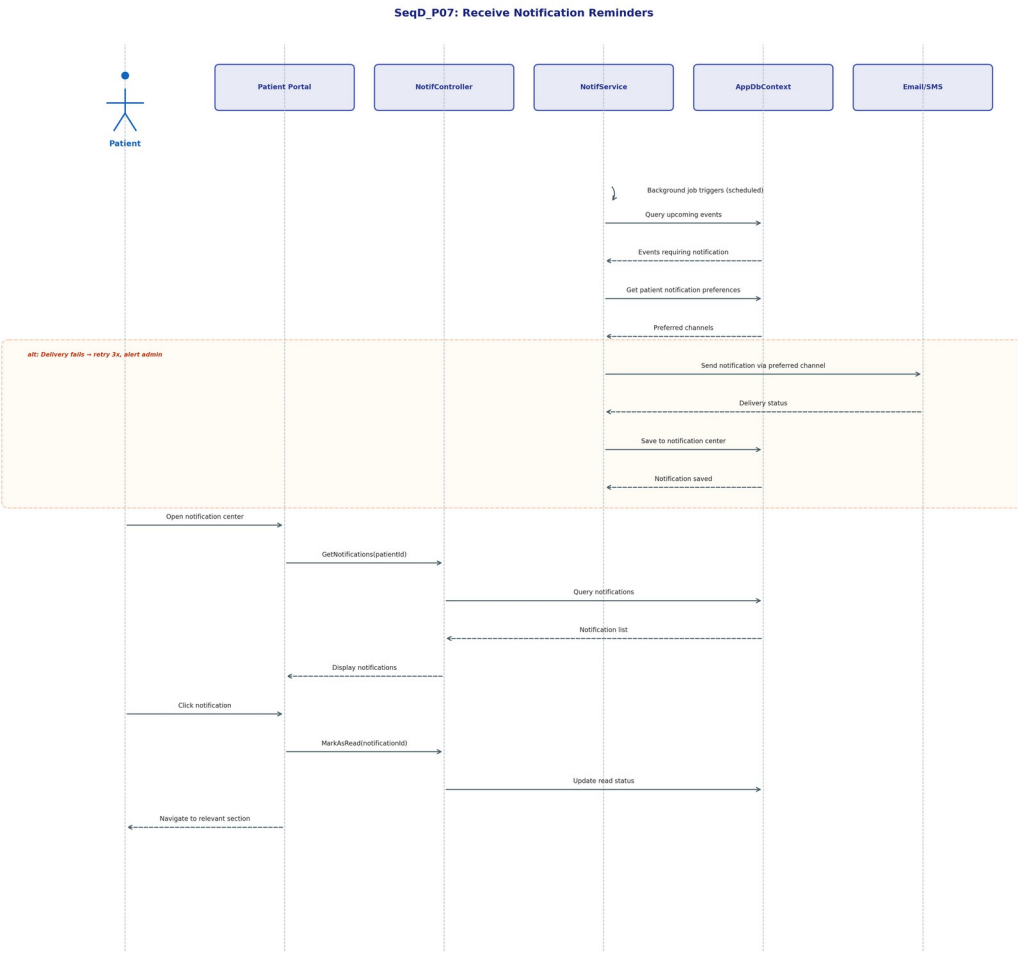
SeqD_P06: Track IVF Cycle Progress

Retrieving cycle data, timeline display, phase details, cycle switching.



SeqD_P07: Receive Notification Reminders

Background job, email/SMS delivery, retry logic, notification center.



SeqD_P08: Submit Feedback

Feedback with appointment linkage, validation, quality management queue.

